

# Santa Cruz ARES Runs Another Successful S.A.F.E. Event

By [Allison Hershey, KM6RMN](#)

Santa Cruz (California) area ARES organizers ran their fourth SAFE (Scavenge Around Field Exercise) on January 9, 2022. Designed in the early days of the pandemic with social distancing in mind, this event allowed participants to get outdoors and communicate with a multi-location incident command team while honing their emergency radio skills.

The first three SAFE events were run in 2020 and early 2021. Any local licensed radio operators interested in emergency preparedness were invited to participate. At event start time, participants checked in to a resource net. Once checked in they were directed to one of two tactical nets to receive a series of assignments to drive to locations and report specific information about them. The observation questions were simple, such as, "what color is the welcome sign at this particular address," and the assessment could be done without leaving the vehicle. Upon completion of their assignments, participants returned to the resource net to demobilize, drove home, and contacted the resource net one more time to report safe arrival. A few field operators' locations were tracked through APRS, bringing a new dimension to the event. This aspect of the exercise will be covered next time, as more participants will learn this technology in the coming year.

Preparation for SAFE IV was started in the fall of 2021. After Santa Cruz ARES members expressed interest in repeating the exercise, Santa Cruz County DEC John Gerhardt, N6QX, recruited volunteers to form a core organizing committee that met online and exchanged emails. Many of the materials were already created in previous events and kept in a shared Google Drive folder: ICS (Incident Command System) Forms 202, 205, 214 for instruction and reporting, vetted participant assignments, scripts, radio protocol pointers, and the all-important Exercise Location and Tracking Sheet. This shared Google spreadsheet was created by JoMarie Faulkerson, KM6URE, to be the "whiteboard" of central operations, displaying the progress of every participant to the core team in their virtual incident command center. Entries made by any team scribe (spreadsheet editor) would be displayed instantly.

There were a dozen planners involved, but the core Incident Command team during the event involved six people. Alex Hays, AJ6QY, and John Kienitz, NS6K, ran the resource net on the WB6ECE 70 cm repeater.

The resource net opened the exercise, checked in radio operators, and directed them to the tactical nets for assignments. At the close of the exercise, they received check-outs and home-safe communications. Bill Tyler, AJ6CQ, and Stephen Betita, KM6NEP, ran Tactical Net A on the K6BJ/KJ6FFP repeater. Bruce Hull, KN6DBR, and JoMarie Faulkerson, KM6URE, ran Tactical Net B on the WR6AOK repeater. Tactical nets fielded subsequent communications

with radio operators, gave and tracked assignments, received and logged reports, transmitted periodic reminders, and solicited wellness reports. Each net partnership consisted of a radio operator and a scribe. It was up to each team as to whether the players stayed in assigned roles or switched roles part way through the exercise. All three teams opted to take turns for a more rounded experience. DEC Gerhardt was present as observer and advisor during the operation, while Allison Hershey, KM6RMN, observed and took notes.

Several features of the event would seem counter to good planning. But in this case, net control operators were being trained to set up emergency nets and field all the problems that might occur in an emergency situation and circumstances. So, there was no prior registration for field participants. (Net traffic and operators being unpredictable in a disruptive event.) Also, any amateur radio operators, not just ARES members, were allowed to take part, simulating the untrained operators rushing to their radios during an emergency.

Field participants were instructed to check in as soon as they could sense an opening at start time. (Net control operators needed to sort out pileups as efficiently as possible.) The secondary tactical nets were not assigned, but chosen at the field operator's discretion, risking crowding on one channel and inactivity on the other. (A one-off experiment to see how the distribution would fall.)

The event was highly organized in other ways, with scripts and instructions honed over the last 2 years. All field assignments were grouped by general location, codified, numbered, had reference photographs, and were easily accessed by the team online. The Exercise Location and Tracking Sheet (a Google spreadsheet refined over several exercises) was technically in two sections: one side filled out by the resource net team, the other by the tactical net teams. Yet its information was instantly available to everyone on the command team. It was constructed in such a way as to require all participant field operators to check-in through the proper channels before they could receive an assignment. Standard ICS forms were also used so that all participants would gain experience in gathering and disseminating information in a way most useful to emergency services.

Check-in time was 1:00 PM. Twenty minutes prior to check-in, the net operators connected from their homes via Zoom in their virtual command center. Exercise frequencies were monitored on home radios.

Though an operator's Zoom mic was muted during transmissions, Gerhardt and the net control operators talked freely between transmissions to correct or instruct as the exercise went on. They practiced using clear language conventions learned in ARES meetings and nets, such as phonetic spelling of names when requested, careful parsing of numbers, keeping messages brief, and using efficient call and response methods with field operators.

The expected pileup was sorted out in the first 20 or 30 minutes, but straggling check-ins continued to the end of the first hour of the two-hour event. Field operations went fairly

smoothly, with a few minor mishaps in execution as expected. After all, it was a training exercise. Eighteen participants checked in, completed tasks, and checked out. Almost all of them remembered to notify the resource net when they arrived home safely, an improvement over previous SAFE events.

The core team learned a few lessons as well. One was that allowing participants to pick their own tactical net without guidance caused extremely lopsided distribution. Net controls on K6BJ fielded most of the calls, while those on WB6AOK had very few. This was due to participants being more confident in one repeater than the other, as well as some reception issues in the area's mountainous terrain.

There were a few cases of perceptual disconnect between apparent map distances and travel time. Tactical net controls had been instructed to make a participant's later assignments close to their initial ones to reduce driving distances. But some locations that appeared close together were difficult to traverse -- the maps did not show uncross-able canyons and gated roads. There was also an APRS issue during the exercise.

The biggest learning opportunity occurred when two radio operators checked in separately while riding in the same vehicle. The tactical net control operator didn't realize this and gave them separate assignments far apart. Not immediately informed of the issue, the command team experienced some confusion about the companion's whereabouts until one net control operator recognized the two were connected. This was discussed afterwards, and suggestions taken how to prevent this kind of misunderstanding in the future.

Two problems they prepared for didn't happen. The net operators kept paper versions of all forms and a hard copy of the Exercise Location and Tracking Sheet, instructed to "use them as best they could" if internet connections were interrupted. In an emergency event, lack of internet access would be a real possibility. Similarly, if a repeater went down, operators were instructed to move to one of the other repeaters and carry on. Neither glitch occurred, which Gerhardt seemed to find mildly disappointing--a lost learning opportunity.

Was the exercise a success? Yes. Was it fun? Yeah! Those in the field enjoyed their adventures exploring Santa Cruz County. Net control operators were pleased to master a few more radio wrangling skills. Organizers improved their lesson plans and added another feather to their collective cap. It looks like SAFE will be a regular event going forward, even as pandemic restrictions ease. *[The author is Santa Cruz County, California Public Information Officer and Assistant Emergency Coordinator]*